DATA ITEM DESCRIPTION

Title: SOFTWARE ASSURANCE CASE

Number: DI-SESS-82409
AMSC Number: N10393
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20230608
Limitation: N/A
GIDEP Applicable: N/A
Project Number: SESS-2023-011

Use/Relationship: The Software Assurance Case is a structured argument, supported by evidence, intended to justify that the software/system is acceptably assured relative to a concern (such as safety or security) in the intended operating environment.

This Data Item Description (DID) contains the format, content, and intended use for the data product resulting from the work task described in the contract Statement of Work/Performance Work Statement.

Requirements:

1. Referenced documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

1.1. National Institute of Standards and Technology IR 7608 -Software Assurance Using Structured Assurance Case Models. https://www.nist.gov/publications/software-assurance-using-structuredassurance-case-models

1.2. "DoD Developer's Guidebook for Software Assurance," Carnegie Mellon University, Software Engineering Institute (SEI), Federally Funded Research and Development Center (FFRDC).
https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=539177

1.3. "Program Manager's Guidebook for Software Assurance," Carnegie Mellon University, Software Engineering Institute (SEI), Federally Funded Research and Development Center (FFRDC).
https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=538771

1.4. Program Protection Plan (PPP).

1.5. ISO/IEC/IEEE 12207, Systems and Software Engineering (or equivalent).

2. Format. Contractor Format.

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3. Content. The documented evidence shall provide a convincing and valid argument that a specified set of critical claims regarding a software's/system's properties are adequately justified for a given application in a given environment. The Software Assurance Case shall be organized into the following sections:

3.1. Reference Documents. This section shall list the number, title, revision, and date of all documents referenced in this deliverable. This section shall also identify the source for all documents not available through normal Government stocking activities.

3.2. Document Management. This section shall identify the version, release date, and other relevant management and configuration control information associated with the current version of the deliverable. A change history, highlighting significant changes from version to version shall be included.

3.3. Table of Contents. This section shall index all sections, major paragraphs, subparagraphs and appendices with page numbers.

3.4. Scope. This section shall be divided into paragraphs covering system and software identification, system overview and document overview.

3.4.1. System and Software Identification. This section shall contain a full identification of the system and the software to which this deliverable applies, including, as applicable, identification number(s), title(s), abbreviation(s), version number(s), and release number(s).

3.4.2. System Overview. This section shall briefly state the purpose of the system to which this deliverable applies. It shall describe the general nature of the system and software; summarize the history of system development, operation, and maintenance; identify the project sponsor, acquirer, user, developer, and support agencies; identify current and planned operating sites; and list other relevant documents.

3.4.3. Document Overview. This section shall summarize the purpose and contents of the deliverable and shall describe any security or privacy considerations associated with its use.

3.5. Software Assurance Case Model. The structured assurance case model is represented as a directed graph whose nodes consist of claims, arguments, and evidence elements.

3.5.1. Claims and Sub-Claims. A claim is a statement asserting some characteristic, property, or behavior of the software or system that can be evaluated for truthfulness, is demonstrable, and is supported by arguments based on objective evidence. A claim may be further decomposed into subclaims, and expressed either as a positive or negative statement.

3.5.1.1. Claims and Sub-Claims should be pre-reviewed to ensure non-duplication.

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3.5.2. Arguments. Arguments are logical propositions intended to support a claim through reasoning or logic that links evidence to a claim. An argument is the explanation of how the evidence can be interpreted as supporting a claim or sub-claim.

3.5.3. Evidence. Evidence is information used to support a claim. Evidence should be objective, reproducible, repeatable, and non-disputable.

End of DI-SESS-82409

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